

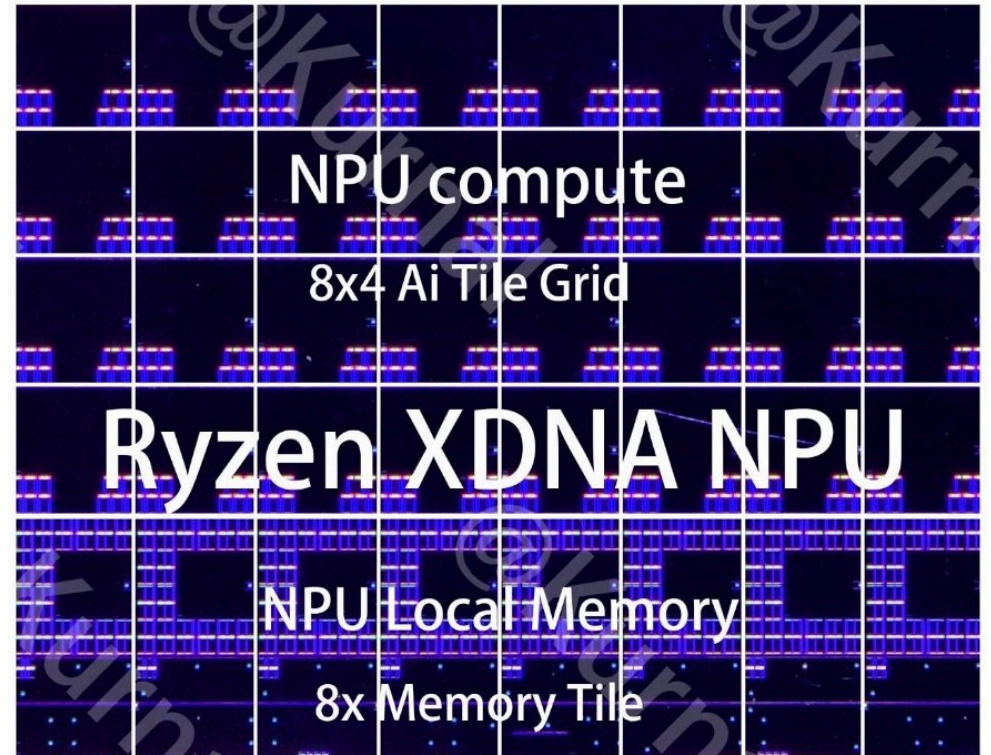
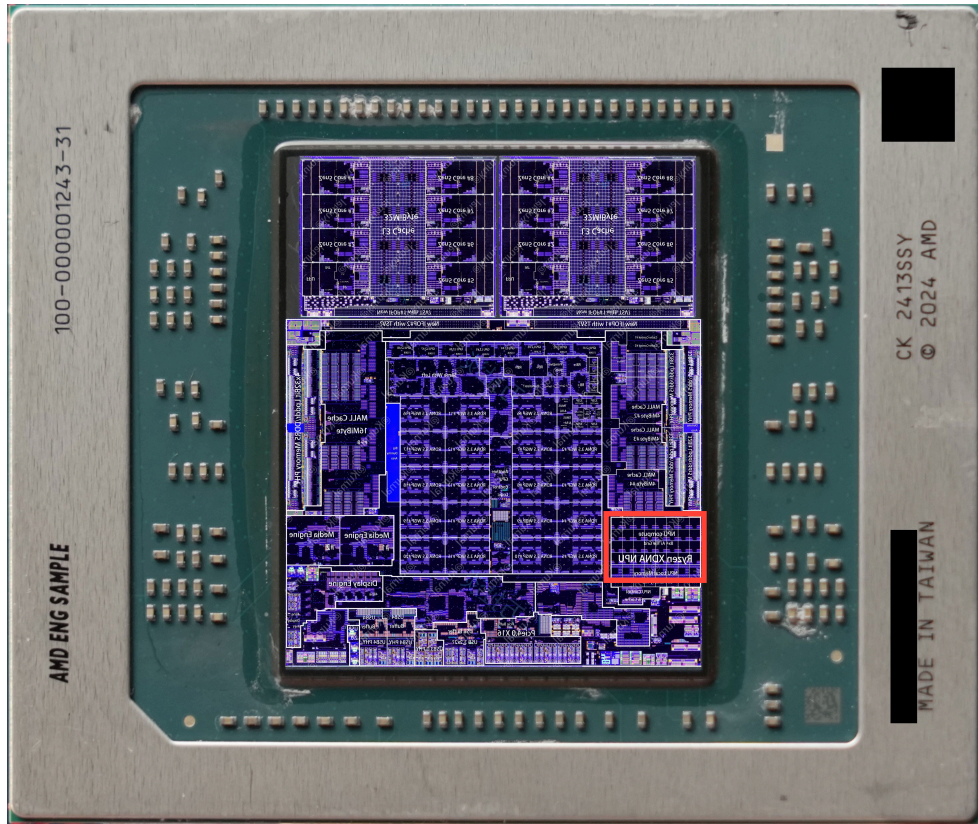


XDNA Architecture

Tamino Steinert

Architecture Overview

Die Shot

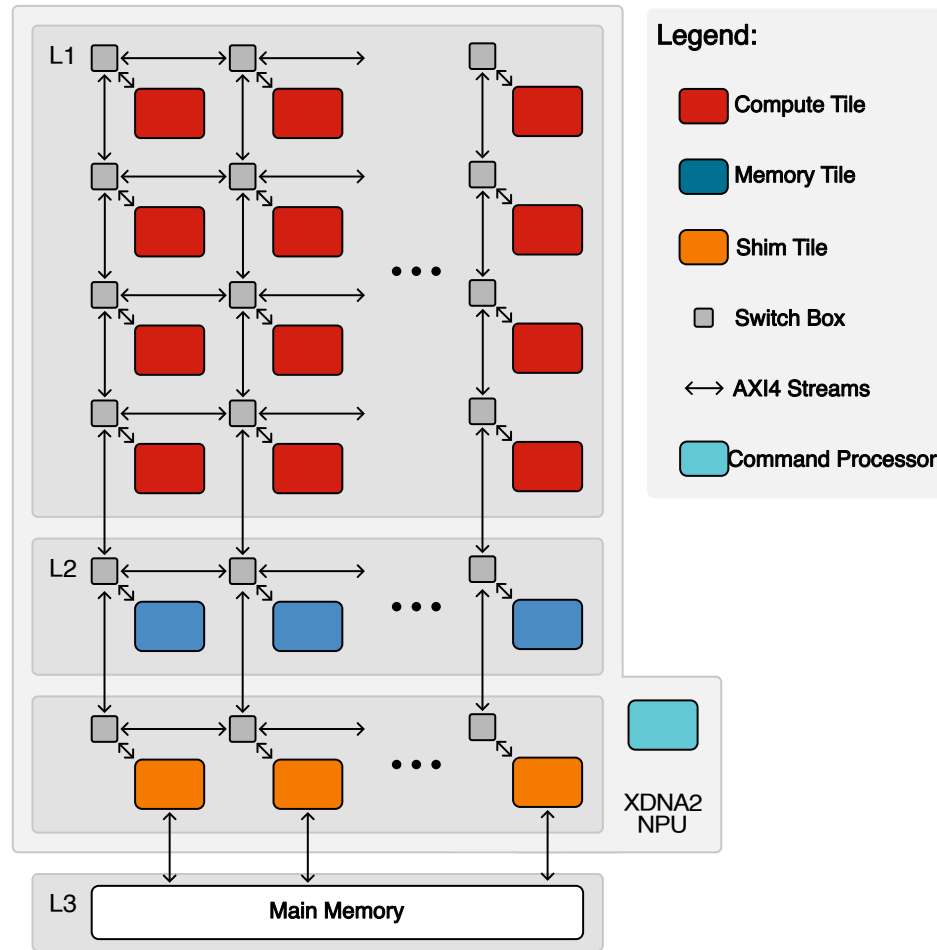


AMD Ryzen AI Max+ PRO 395

<https://x.com/Kurnalsalts/status/1891873628737896680>

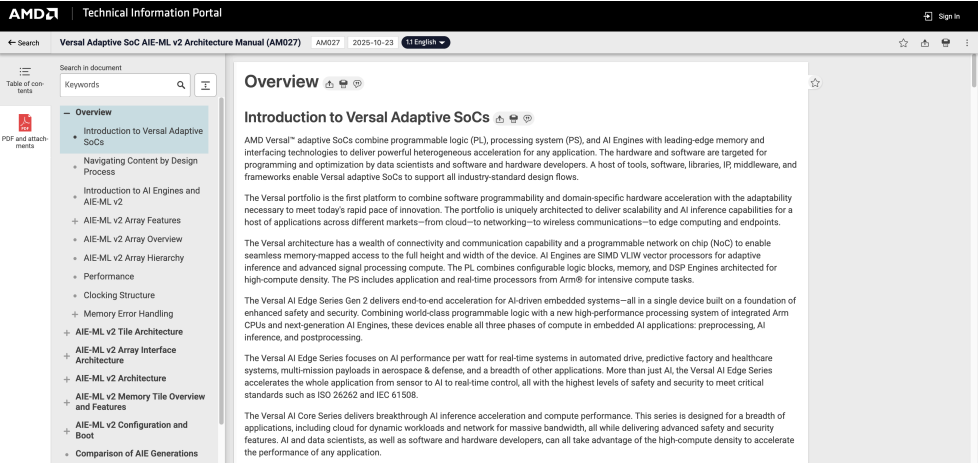
Overview

- Consists of three tile types: compute, memory, and shim.
- 2D layout with 8 columns and 6 rows:
 - Row 0: shim tiles
 - Row 1: memory tiles
 - Row 2–5: compute tiles
- Compute and memory tiles have scratchpad memory.
- Data transfer between tiles through network-on-chip (NoC) and direct memory access (DMA).

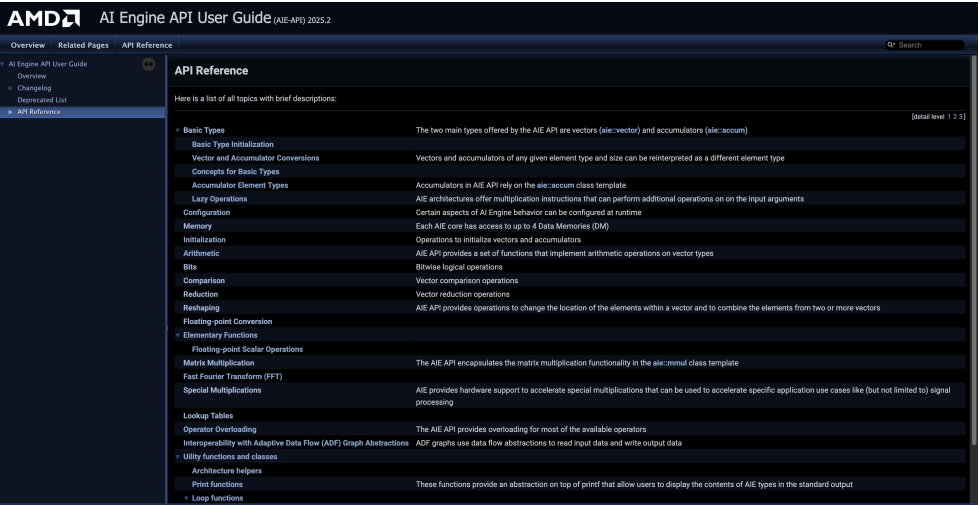


XDNA2 Architecture Overview

Documentation



AMD – AIE-ML v2 Architecture Manual – AM027



AMD – AI Engine API User Guide – AIE-API